



OPEN Effects of temporal, geographical and environmental factors on salmon lice (*Lepeophtheirus salmonis*) levels of Atlantic salmon (*Salmo salar*) in Ireland

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The salmon louse (*Lepeophtheirus salmonis*) poses a major challenge for Atlantic salmon (*Salmo salar*) aquaculture in the northern hemisphere. These ectoparasites feed on the blood, tissue, and mucus of their fish host, causing skin damage, increasing susceptibility to secondary infections, and, in severe cases, leading to fish mortality. The economic impact on aquaculture is substantial due to treatment and management costs. Effective management strategies rely on a comprehensive understanding of infection dynamics. To investigate these dynamics, we analysed a 30-year dataset on sea lice abundance from Atlantic salmon farms in Ireland, collected by the Marine Institute through the National Sea Lice Monitoring Programme. Using generalised linear mixed-effects models, we detected temporal fluctuations, with an overall decline in lice abundance. Seasonal variation was evident, with the lowest abundance in spring and the highest in autumn. Sea lice abundance was influenced by farm exposure, represented by bay mouth width, bay length, number of active farms per bay and dissolved oxygen levels. By identifying key factors driving infection patterns, this study provides valuable insights for predicting high-risk periods and optimising management strategies. These findings will inform strategies for salmon lice control, contributing to improved fish health and sustainability in aquaculture.

Keywords Epidemiology, Aquaculture, Long-term changes, Parasite, Fish health

One of the main challenges for salmon aquaculture is the management of ectoparasitic sea lice¹. In the Northern Atlantic, the primary sea louse species affecting farmed salmon is *Lepeophtheirus salmonis*, while other species, such as *Caligus rogercresceyi* and *C. teres*, have a greater impact in regions like Chile^{2–7}. Sea lice infections impose increased costs on the industry, not only by negatively affecting fish welfare⁸, but also through the expenses associated with treatment and management strategies¹. Moreover, sea lice treatments have been associated with declining fish welfare, compounding economic losses^{9,10}.

Various management strategies are implemented to maintain low infection levels of sea lice on salmon farms, from husbandry practices to veterinary medicinal and non-medicinal treatments^{8,11–14}. Integral to these efforts are monitoring programmes that ensure the effectiveness of treatments^{15,16}. In Ireland, the National Sea Lice Monitoring Programme, overseen by the Marine Institute¹⁷, was introduced in 1991 to monitor sea lice in the salmon farming industry. As of 2023, the industry consists of 20 marine farms located along the west coast, producing 9300 tonnes of Atlantic salmon^{17,18}.

To successfully implement effective management strategies, it is necessary to understand the dynamics of sea lice infections on salmon farms^{19,20}. Although several previous epidemiological studies have sought to identify factors influencing the abundance of *L. salmonis* on Atlantic salmon farms¹⁹, to date, comprehensive epidemiological studies are limited^{3,19,21,22}. Several studies have investigated various aspects of the epidemiology of sea lice on Atlantic salmon farms in Ireland, however, there is no recent comprehensive study on this topic^{23–25}.

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From previous studies investigating infection levels of *L. salmonis* on salmon farms a range of influential factors were described including: environmental factors (e.g. salinity, temperature and water circulation)^{19,26,27}; geographical factors (e.g. location of farm, vicinity to other farms)^{19,28}; demographic factors^{19,29}; and husbandry practices (e.g. stocking density, species of fish, treatment)¹⁹. Husbandry practices are especially important to consider as they can be employed as mitigating strategies by the salmon farm industry. For example, controlling stocking density can be used to prevent sea lice outbreaks regionally³⁰. However, since the dynamics of sea lice infections are influenced by a complex interaction of factors, examining parameters beyond the control of salmon producers is also essential to better understand variation in sea lice levels^{31,32}.

Assessing the factors that influence sea lice infection dynamics can aid in predicting periods of higher risk of infection and in implementing successful management strategies to reduce parasite burdens and maintain fish health, both in aquaculture¹⁹ and in wild salmon populations³³. The aim of this study was to assess the effects of temporal, geographical and environmental factors on long-term changes in sea lice abundance on Atlantic salmon farms in Ireland. Sea lice infection dynamics over a 30-year period were examined. The effect of geographical factors describing the location of salmon farms (depth, distance from mouth of bay, width of bay, length of bay, number of active farms per bay), and environmental factors (temperature, salinity, dissolved oxygen), on sea lice abundance were determined.

Methods

Sea lice data

Sea lice count data were collected from all active salmon farms along the west coast of Ireland, provided by the Marine Institute as part of the National Sea Lice Monitoring Programme. Monitoring of sea lice takes place 14 times a year: twice a month in spring (March, April, May), once in winter (December or January) and once a month throughout the rest of the year. At each inspection, two pens are selected—one standard pen that is inspected each time and one random pen chosen opportunistically. In each selected pen 30 fish are anaesthetised (using tricaine methanesulfonate in seawater) and examined for sea lice. Counts of all sea lice life stages are recorded, and all mobile life stages are collected in 70% ethanol for confirmation of correct identification. The water with anaesthetic is sieved for detached sea lice. The mean number of lice per pen is calculated including the number of detached sea lice in sieved water samples. All active salmon farms are monitored, the farms are not active between separate stocks of fish when the farm is left to fallow¹⁷.

The work presented here is based on two complementary datasets: a 30-year dataset (1993–2023) with sea lice abundance recorded at pen level, and an 18-year dataset (2006–2023) with records at individual fish level. Each of the datasets was used to answer different questions, the 30-year dataset was analysed to determine long-term and seasonal trends in sea lice abundance, and the 18-year dataset was analysed to determine the variation in sea lice abundance with geographical and environmental factors. The datasets overlap but differ only in the level at which sea lice abundance was recorded and the absence of detached and sieved sea lice data in the 18-year dataset. There are two gaps in the data due to the Covid-19 pandemic restrictions (2020: April to June; 2021: February to April), when the Marine Institute did not carry out sea lice monitoring on salmon farms. Instead, the salmon producers were instructed to carry out their own inspections and report back mean values per pen^{25,34}. The reported values by the producers were incorporated only into the 30-year dataset to fill in the gaps for these years, while these gaps remain in the 18-year dataset.

Long-term trends in sea lice abundance

The 30-year dataset consists of sea lice counts recorded as mean value per pen (recorded for two pens during each farm inspection) and spans from 1993 to 2023. The original raw data contains 16,955 mean values collected from 67 salmon farms. Outliers were identified and discarded, including salmon farms with fewer than 120 recorded sea lice pen-level inspections—excluded after initial exploration showing that datasets with too few samplings were not suitable for robust analysis—and samplings from fish that spent more than 24 months in seawater. In addition, only data from full production cycles of fish were kept, leading to the exclusion of fish stock transported to sea water in 2023. The final dataset contains 13,484 mean values that were sampled within 28 salmon farms. To account for the total population of sea lice (*L. salmonis*) the overall abundance of sea lice used in our main analysis of the 30-year dataset focuses on the mean number of total mobile sea lice (adults and preadults). The abundance of either total adult female lice or ovigerous female lice (adult females with egg strings) is used in different countries to trigger treatments for sea lice on salmon farms^{35–38}. From the 30-year dataset, data were also available for ovigerous female sea lice and additional analyses based on these data are provided in the Supplementary Information.

A list of factors possibly impacting sea lice abundance was identified based on current knowledge^{19,26–29} and expert consultation with the Marine Institute. These factors were: date of sampling (year and month), farm location (pen name-standard and random, farm name, latitude, longitude), class of fish (season when smolts were transported to seawater), and duration of time fish spent in seawater prior to sampling. The year of sampling was used both as a numeric and categorical variable to determine the overall trend in sea lice abundance as well as changes between individual years. The month of sampling was recorded in 14 levels according to the time of sea lice inspections (twice in spring, once in December or January, once per month for the rest of the year). The date of each sea lice datapoint was recorded and used to explore variation in sea lice levels with year and month. The time fish spent in seawater before each inspection was calculated using fish transfer data from the Marine Institute Fish Health Unit. Since 2006, fish movement records have been required under Fish Health Regulations, allowing the identification of seawater transfer dates. However, as transfer data have been available only since 2006, many fish stocks lacked specific transfer information. To address this, an average transfer date was calculated separately for fish transferred to sea from hatchery facilities in spring (spring class) and autumn (autumn class). Based on known transfer dates, the mean transfer dates were set to as March 17th for the spring

class and October 31st for the autumn class. Since exact transfer dates were unavailable for most fish stocks, these mean dates were used as approximations for all stocks of fish. Using the estimated transfer dates and sea lice inspection dates, the time fish spent in seawater was estimated. The time in seawater was recorded in months and categorised into four groups: up to 6, 12, 18, and 24 months in seawater.

Variation in sea lice abundance with geographical and environmental factors

The 18-year dataset consists of sea lice counts at fish level recorded separately for each examined fish spanning the period from 2006 to 2023. The original raw data contains 186,796 sea lice counts per fish collected from 41 salmon farms. Salmon farms that had less than 60 sampling points during this time were discarded, as well as samples from fish that spent more than 24 months in seawater. Data with more than six active salmon farms per bay were excluded due to a low number of counts per fish (180 data points). In addition, only data from full production cycles of fish were kept, leading to the exclusion of fish stock transported to seawater in 2023. The final dataset consists of sea lice counts (*L. salmonis*) from 167,533 fish that were sampled within 27 salmon farms. The main analysis of the 18-year dataset focused on the count of total mobile sea lice (adults and preadults). From the 18-year dataset, data were also available for ovigerous female sea lice (adult females with egg strings) and total adult female sea lice (adult and preadult females) and analyses based on these data are provided in the Supplementary Information.

The factors identified for the 30-year dataset were also included into the analyses of the 18-year dataset. These included the date of sampling (year and month), farm location (pen name, latitude and longitude), fish class, and duration of time fish spent in seawater. Both month and year were included as numeric predictors to account for the overall change in sea lice counts. The aquaculture company variable was divided into three levels, two for the two salmon companies owning the highest number of salmon farms over the last 30 years and the third including all other companies. Time in seawater was included as for the 30-year dataset. The exact date of transfer was possible to determine for only 45% of fish stocks in the 18-year dataset due to transfers between farms during the production cycle, which introduced uncertainty in tracking the original transfer dates. Therefore, the approximated mean date of transport was used in cases of missing data on the exact date of transport. The length of fallow periods (time between the end of one production cycle and start of the next one) was calculated as the number of months between the last inspection of the previous fish stock (as an approximation of harvest date) and the first inspection of the new fish stock (as an approximation of fish transfer to seawater). For farms that were inactive for extended periods, any calculated fallow period longer than 24 months was replaced with a value of 25 months ($\mu = 8$ months, $\sigma = 7$ months). This approach allowed us to account for prolonged fallow times while avoiding bias associated with farm inactivity.

In addition, geographical and environmental factors of interest, available for analysis of the 18-year dataset, were included. To describe the geography of the location of salmon farms and of individual bays several different variables were defined: depth, length of bay, width of mouth of bay, distance of farm from mouth of bay and the number of active farms per bay. The depth of water at the location of each farm ($\mu = 17.3$ m, $\sigma = 7.1$ m) was established from the INFOMAR dataset (Table 1), and the geographical factors relating to each bay were determined using the coordinates of each farm and the geography of each bay (Fig. 1). The factors recorded were the length of the bay ($\mu = 20.7$ km, $\sigma = 12$ km), width of the mouth of bay ($\mu = 7.9$ km, $\sigma = 3.9$ km), and distance

Variable	Provider	Name of dataset	Citations
Depth of water	Integrated mapping for the sustainable development of Ireland's marine resource (INFOMAR)	Barometry data	39
Geographical characteristics of bay	Google earth		40
Temperature	Marine institute	Sea lice database	
Temperature	Environmental research division's data access programme (ERDDAP), Marine Institute	Model monthly means	41
		Ireland's Marine Institute research vessel underway data 2013–2018	42
		Coastal temperature network	43
	Copernicus marine environment monitoring service (CMEMS)	Global ocean OSTIA sea surface temperature and sea ice analysis	44
Salinity	Marine Institute	Sea lice database	
Salinity	ERDDAP, Marine Institute	Ireland's Marine Institute research vessel underway data 2013–2018	42
	CMEMS	Global ocean biogeochemistry analysis and forecast	45
		Global ocean physics reanalysis	46
		Global ocean physics analysis and forecast	47
Dissolved oxygen	Marine Institute	Sea lice database	
Dissolved oxygen	CMEMS	Global ocean biogeochemistry analysis and forecast	45
		Global ocean biogeochemistry hindcast	48
		Atlantic—European northwest shelf—ocean biogeochemistry analysis and forecast	49
		Atlantic—European northwest shelf—ocean biogeochemistry reanalysis	50

Table 1. Resources used to obtain data on the geographical and environmental variables relating to each active salmon farm or bay in the 18-year dataset.

of each farm from the mouth of the bay ($\mu=6.7$ km, $\sigma=7.2$ km). The distance of the farm from the mouth of the bay was measured as the approximate sea-way distance. Recorded negative values for the distance of the farm from the mouth of the bay signify that the farm is located outside of the bay. The number of active farms per bay was determined for each month separately ($\mu=1.7$, $\sigma=0.9$).

Data for three environmental variables, temperature, salinity and dissolved oxygen, were obtained from several online sources (Table 1). More than one data source was needed to cover the full time period and all locations. When more than one source was available for the same datapoint, the data with better accuracy (measured at the farm location and/or on the inspection date) were selected; otherwise, a mean of observed values was used. The data were recorded for each bay where salmon farms were located and each month when the farms were active in the individual dataset. All environmental variables are surface measurements collected from one metre under the surface. Surface water temperature ranged from 3 to 21 °C ($\mu=11.6$ °C, $\sigma=3$ °C); the highest temperatures were recorded in October 2020; the coldest water temperature was recorded in February 2019. Salinity ranged from 23.2 to 37 ppt ($\mu=33.5$ ppt, $\sigma=2.4$ ppt) and dissolved oxygen ranged from 73.5 to 125% ($\mu=96.9\%$, $\sigma=8.4\%$).

Statistical analysis

To determine the effects of temporal, geographical, and environmental factors on sea lice abundance detected on salmon farms in Ireland, two separate Generalised Linear Mixed-Effect Models (GLMMs) were implemented. As the response variables included both mean values of sea lice abundance and count data, and repeated measurements were taken from the same farms over time, GLMMs provided a flexible framework to handle overdispersion, non-independence, and random variation among farms⁵¹.

The models were constructed to analyse the data at several different levels and capture variation in sea lice abundance. The 30-year dataset was analysed using the mean sea lice abundance at pen and farm level, while the 18-year dataset was analysed using the sea lice count at fish level and the estimated mean sea lice abundance at pen and farm level. Since the sea lice abundance at pen and farm level from both the 30- and 18-year datasets were recorded or estimated as mean values and exhibited a suitable distribution, they were modelled using a gamma distribution with a log-link function and a zero-inflation term to account for presence of zero counts. Additionally, sea lice counts at fish level from the 18-year dataset, recorded as a count of sea lice per fish, were modelled using a negative binomial distribution to address high overdispersion ($\theta=1.7$).

The residuals from each model were tested for temporal autocorrelation using autocorrelation plots and tests for detection of autocorrelation (Durbin-Watson (dw) statistics⁵², Breusch-Godfrey test^{53,54}). The fit of models to data at pen and farm level for the 30-year dataset and on data at fish, pen and farm level for the 18-year dataset were compared. The model for the 30-year dataset analysed at pen level showed reduced autocorrelation (year categorical: dw = 1.94, year continuous: dw = 1.94) compared to the models analysed at farm level (year categorical: dw = 1.84, year continuous: dw = 1.77). On the other hand, the model for the 18-year dataset analysed at farm level showed reduced autocorrelation (dw = 1.45) compared to the models analysed at fish (dw = 0.91) and pen level (dw = 1.14). Based on statistical diagnostics, data availability, and the research question the pen level data for the 30-year dataset and the farm level data for the 18-year dataset were used for further analysis.

To assess potential collinearity among predictors, a multicollinearity test was performed. Multicollinearity was detected only in the 18-year dataset, specifically among bay length, latitude, and longitude. This issue was

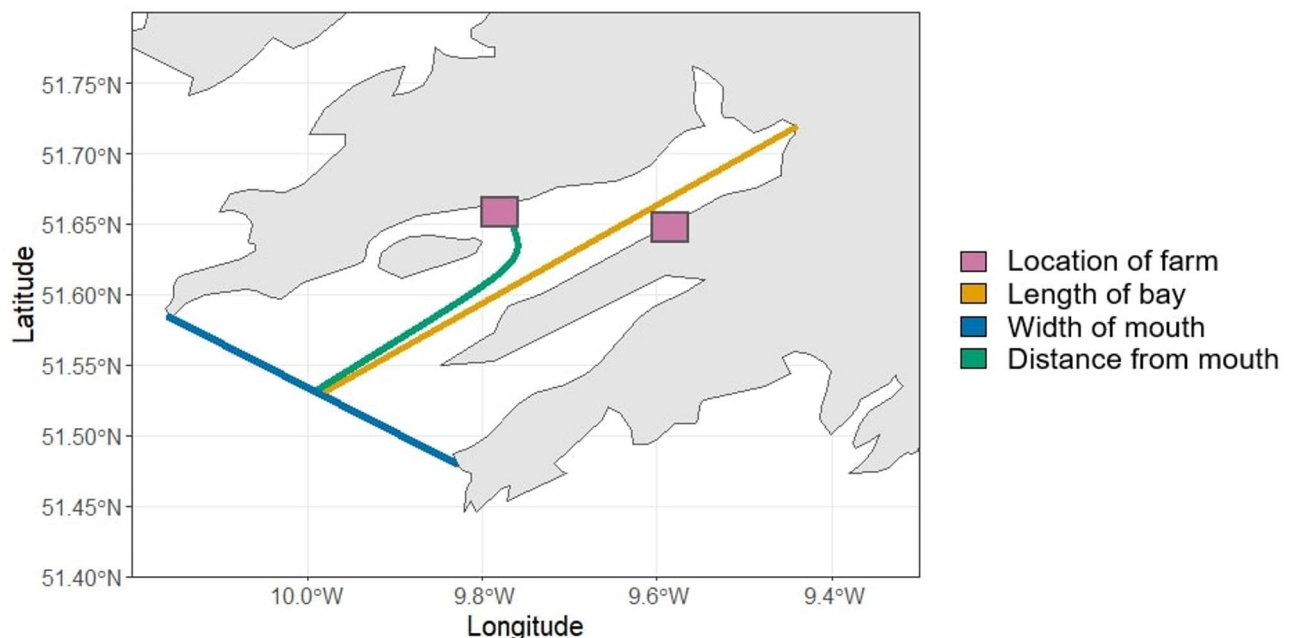


Fig. 1. Map of Bantry Bay illustrating the measurement of geographical factors.

resolved by eliminating longitude as a predictor. Based on the initial analyses of the data, the final models were tested starting with a full model including all candidate variables and using a backward elimination based on Akaike Information Criterion (AIC). The suitable fit of each model to the data was also assessed using the Bayesian Information Criterion (BIC) model selection method, alongside the percentage of residual deviance explained.

The two final models for the 30-year dataset included the random effect of farm to account for the non-independent observations. No random effect was included in the final model for the 18-year dataset, because the geographical factors were recorded either on the level of farm or on the level of bay, accounting for the variation between farms. All continuous predictor variables were centered around the mean.

The marginal effect of each predictor variable was assessed using a Type II Wald Chi-square test⁵⁵, which evaluates the significance of each predictor while accounting for the effects of the other variables in the model. Statistical significance was determined at P value of <0.05 . All statistical analyses were carried out using the statistical computing environment R 4.3.3⁵⁶. The package 'glmmTMB'⁵⁷ was used to test models, 'ggplot2'⁵⁸ was used for results visualisation, and 'rnatualearth'⁵⁹ was used for map visualisation.

Results

Long-term trends in sea lice abundance

The 30-year dataset was analysed to determine long-term trends in sea lice (*L. salmonis*) abundance. In total the dataset contained 13,484 mean values of total mobile sea lice at pen level ($\mu = 3.88$, $\sigma = 8.03$), (Fig. 2a). The final models included the random effect of farm and fixed effects of year, month, time in sea, class, latitude, longitude and pen in the case of the model with year as a continuous variable.

Statistical models with year both as a categorical and continuous variable were used to determine the changes between individual years as well as the overall trend in sea lice abundance. The final models examining total mobile sea lice explained 48% and 42% of the overall variation in sea lice abundance for the categorical (AIC 55,088, BIC 55,868) and continuous (AIC 56,314, BIC 56,659) year models, respectively. All predictor variables

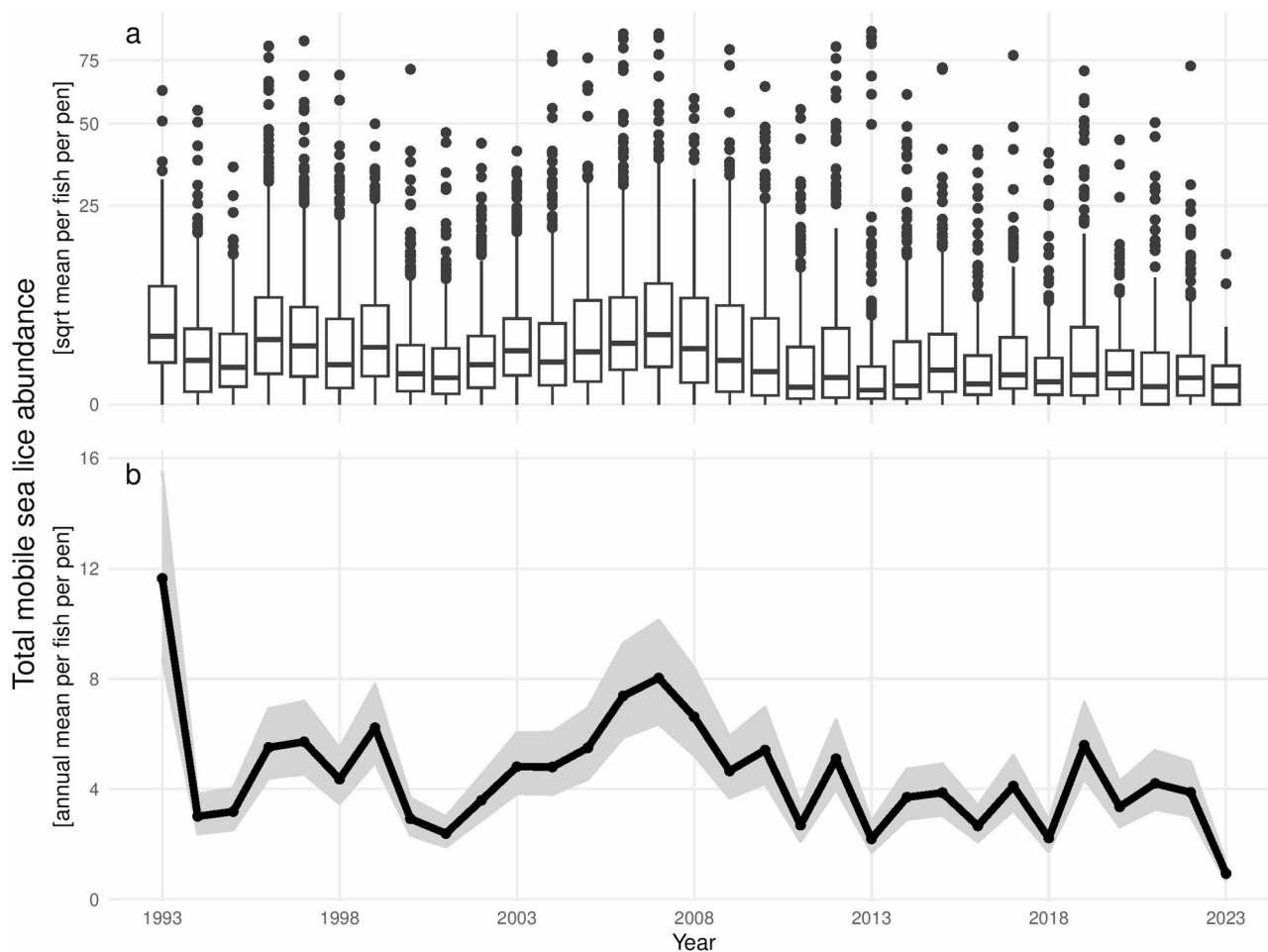


Fig. 2. (a) Detected abundance of total mobile sea lice (adults and preadults) recorded as a mean per pen on all salmon farms over a 30-year period, (b) predicted abundance of total mobile sea lice (adults and preadults) recorded over a 30-year period as an annual mean per pen (black line). The modelled 95% confidence interval is highlighted in grey.

(a) Year categorical			(b) Year continuous		
Predictors	χ^2	p value	Predictors	χ^2	p value
Year	1054	<0.01	Year	47	<0.01
Month	920.7	<0.01	Month	764.4	<0.01
Time in sea	3090.5	<0.01	Time in sea	2921.3	<0.01
Class	29.9	<0.01	Class	21.7	<0.01
Latitude	15.1	<0.01	Pen	5.2	0.02
Longitude	4.3	<0.04	Latitude	17.3	<0.01
			Longitude	5.6	0.02

Table 2. Results of type II Wald chi-square test of generalised linear mixed-effect models examining the abundance of total mobile sea lice recorded in the 30-year dataset as a mean value per pen in relation to temporal factors (year, month). The results are shown for both final models (a) year categorical and (b) year continuous predictor.

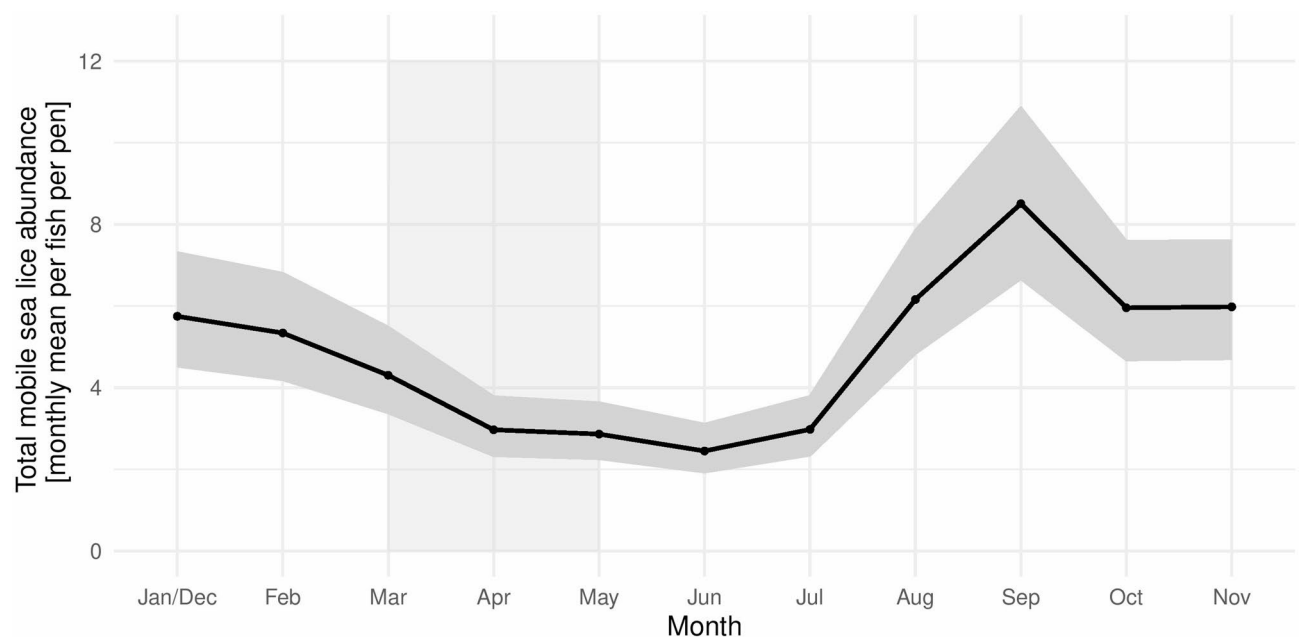


Fig. 3. Predicted abundance of total mobile sea lice (adults and preadults) recorded over months as a monthly mean per pen (black line) for the model with year as a categorical variable for the 30-year dataset. The modelled 95% confidence interval is highlighted in dark grey. Stricter management strategies are implemented on salmon farms during spring (March to May)¹⁷, highlighted in light grey.

were detected to have significant effects on sea lice abundance of total mobile sea lice (Table 2; Supplementary Information). The abundance of total mobile sea lice showed cyclic fluctuations across years (Fig. 2b) and an overall decreasing trend was detected in the data ($Z = -6.86$, $p = <0.01$; Supplementary Information). Peak sea lice abundance levels were recorded in 1993 and 2007, with relatively high variability between these years, ranging from 1.9 to 7.5 total mobile adults per pen. The lowest abundance levels were recorded in 2018 and 2023. From 2011 onwards, the overall magnitude in annual abundance was lower ranging between 1.6 to 4.6 total mobiles per pen, except in 2023, when the abundance dropped to the lowest value of 0.8 (Fig. 2a). Seasonal differences were also detected with a significant overall effect of month ($\chi^2 = 920.7$, $p = <0.01$; Table 2), with a decrease in sea lice abundance at the beginning of spring and beginning of summer. However, a steep increase was observed at the end of summer, with a seasonal peak in September (Fig. 3).

Variation in sea lice abundance with geographical and environmental factors

The 18-year dataset was analysed to determine the variation in sea lice (*L. salmonis*) abundance with geographical and environmental factors. The dataset included 2,993 mean values of total mobile sea lice per farm ($\mu = 3.21$, $\sigma = 7.67$). The final model included fixed effects of year, month, time in seawater, class, company, fallow, latitude, geographical factors (depth, distance of farm from mouth, length of bay, width of mouth of bay, number of active farms in bay) and environmental factors (temperature, dissolved oxygen).

The final model (AIC: 10,397, BIC: 10,619) explained 55% of the overall variation in sea lice abundance for total mobile sea lice. No significant effects on variation in total mobile sea lice abundance were detected for month and class, the geographical variables of depth and distance of farm from mouth of bay (Table 3; Supplementary Information). All other predictor variables (year, time in sea, company, fallow period, latitude) were detected to have significant effects on sea lice abundance of total mobile sea lice (Table 3; Fig. 4). A negative relationship was found between the abundance of total mobile sea lice and the length of bay ($Z = -9.7, p < 0.01$), width of mouth of bay ($Z = -4.4, p < 0.01$), temperature ($Z = -2, p = 0.04$) and the amount of dissolved oxygen in sea water ($Z = -5.2, p < 0.01$), (Supplementary Information). A positive relationship was found between the abundance of total mobile sea lice and the number of active farms per bay ($Z = 4.1, p < 0.01$), (Supplementary Information).

Discussion

Understanding sea lice population dynamics is essential for implementing effective management strategies on Atlantic salmon farms. Here we examined long-term changes in sea lice (*L. salmonis*) abundance at Irish marine salmon farms, considering temporal, geographical, and environmental factors. An overall decline in sea lice abundance was observed, with significant influences from seasonality, geographical factors, and dissolved oxygen levels. The results can be categorised into three groups based on their effects on sea lice abundance: temporal changes, seasonal variation, and farm exposure.

Long-term trends in sea lice abundance

The 30-year dataset offers valuable insights into long-term trends in sea lice abundance on salmon farms in Ireland. Total mobile sea lice abundance showed cyclic fluctuations over the years with an overall decreasing trend, as indicated by the progressively lower magnitude of these fluctuations. A general pattern of higher abundance in one year followed by lower abundance in the next was observed, with a notable exception: a sharp increase from 2001 to 2007, followed by a decline over the next two years. The particularly high abundance recorded in 1993 and the lowest abundance observed in 2023 may be partly influenced by the smaller number of samples available for those years.

An overall reduction of sea lice abundance levels was also observed in a long-term study from Scotland, where data from more than 50 salmon farms recorded over a 10-year period (1996–2006) were analysed⁶⁰. The observed reduction was attributed to improved management strategies, including trained and qualified staff, strategic spring treatments, and the availability of more effective treatments such as emamectin benzoate (SLICE)⁶⁰, although its efficacy declined over time⁶¹. Likewise, a nine-year study in Chile (2010–2018) reported a decreasing trend in sea lice abundance, attributed to the implementation of an official national surveillance and control programme, which included coordinated treatments, early harvest of heavily infected fish, and fallow periods⁶².

Atlantic salmon aquaculture in Ireland experienced steady growth from 1993 to 2001, reaching a peak production of 23,000 tonnes^{35,63–65}. However, this upward trend shifted, and production fluctuated around 12,000 tonnes per year in the following years^{66,67}. Another peak occurred in 2017, with production nearing 20,000 tonnes⁶⁷. In subsequent years, output declined, stabilising at approximately 11,000 tonnes per year, with one of the lowest productions over the last 30-years recorded in 2023 at 9300 tonnes^{18,68}. Salmon production tonnage in Ireland can be used as an indicator of the population of farmed salmon in Irish waters. While higher salmon densities are generally linked to increased sea lice transmission⁶⁹, the highest production levels did not

Type	Predictor	χ^2	p value
Geographical factors	Depth of farm	0.4	0.55
	Distance to bay mouth	2.1	0.15
	Length of bay	93.7	< 0.01
	Width of bay mouth	19.4	< 0.01
	Number of active farms per bay	16.6	< 0.01
Environmental factors	Temperature	4.2	0.04
	Oxygen	27.1	< 0.01
Other factors	Year	54.8	< 0.01
	Month	0.9	0.34
	Time in sea	744.5	< 0.01
	Class	2.5	0.11
	Company	23.08	< 0.01
	Fallow period	7.2	< 0.01
	Latitude	32.1	< 0.01

Table 3. Results of type II Wald Chi-square test of generalised linear mixed-effect model of total mobile sea lice abundance recorded in the 18-year dataset as a mean value per farm in relationship to geographical factors (depth, distance from mouth, length of bay, width of mouth, number of active farms), environmental factors (temperature, dissolved oxygen), and other study factors (year, month, time in seawater, class, company, fallow period, latitude) recorded as part of an 18-year dataset by the Marine Institute.

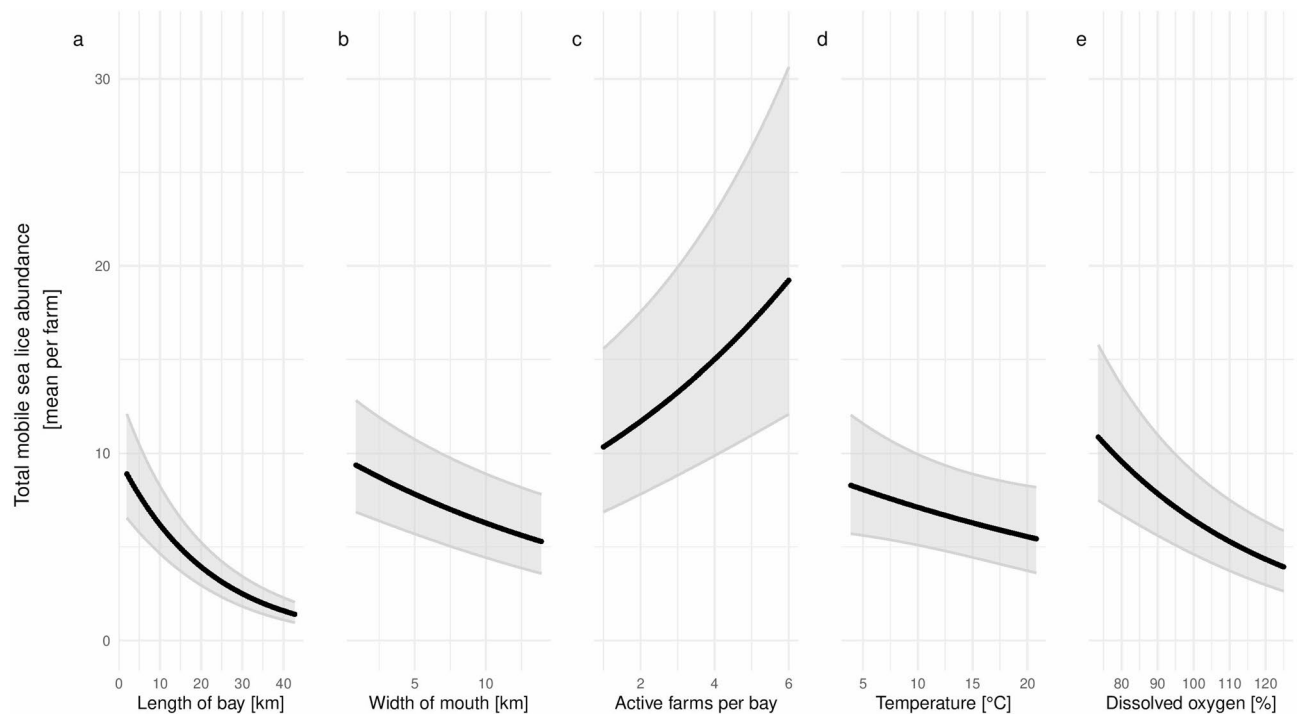


Fig. 4. Predicted abundance of total mobile sea lice (adults and preadults) recorded as part of the 18-year dataset (2006–2023) recorded as a mean value per farm, visualised as mean value (black line) calculated for each of the following variables: (a) length of bay, (b) width of mouth of bay, (c) number of active farms per bay, (d) temperature, and (e) dissolved oxygen. The modelled 95% confidence interval is highlighted in grey.

coincide with the peak sea lice abundances observed in this study (1993 and 2007). In fact, 2007 had one of the lowest production levels in the 30-year period. On the other hand, the lowest sea lice abundance recorded in 2023 also occurred during a year of low production, while a production increase in 1999 corresponded with a rise in sea lice abundance.

Within Ireland, management strategies have changed over the years following the establishment of the National Sea Lice Monitoring Programme in 1991³⁵. In 1993, the Single Bay Management (SBM) framework was first introduced. This framework aimed to enhance treatment coordination within bays, ensure single generation fish farming, and implement fallow periods between fish stocks²³. Further advancements followed in 2000 with the formal publication of the sea lice monitoring protocol⁷⁰. In response to ongoing challenges in maintaining low sea lice abundance, a new pest management strategy was introduced in 2008, strengthening SBM strategies and providing additional recommendations for improved sea lice control³⁵.

Over time, management strategies and treatment availability have evolved considerably. In 2000, the use of ivermectin was discontinued following the full authorisation of an alternative avermectin treatment, emamectin benzoate (SLICE). Ivermectin was phased out due to its narrow safety margin for salmon and its negative impact on benthic communities^{23,71–73}. SLICE has been used for preventative treatments, particularly at the start of spring, a practice adopted in 2000²³. In addition to medicinal treatments, various non-medicinal approaches—such as preventive measures (e.g. bubble curtains, skirts), physical removal (e.g. freshwater treatments), and biological control using cleaner fish—have been implemented^{12,14,74,75}. The use of these methods has become increasingly widespread worldwide in recent years^{8,76–78}. While cleaner fish had been used previously, their widespread adoption in Ireland only began in 2015⁷⁶. Today, cleaner fish are used year-round on most salmon farms in Ireland¹⁷; however, their cost-effectiveness as a control measure against sea lice remains uncertain^{79–81}.

The observed fluctuations and overall decline in sea lice abundance in Ireland over the past 30 years are likely influenced by a combination of these factors, including reduced salmon production, evolving management strategies, available treatment and environmental changes^{18,35,67,82}. However, due to the absence of detailed farm-level and individual fish cohort data for these factors, we could not directly assess the relative impact of the variables. To better understand the drivers behind sea lice abundance trends, further research integrating comprehensive production, management, and environmental data is essential.

Seasonal changes in sea lice abundance

Previous studies have shown that warmer sea temperatures shorten the generation time of sea lice⁸² and lead to the production of larger eggs by females⁸³. Laboratory research has consistently demonstrated a positive correlation between increased temperature and sea lice abundance^{82–84}. However, epidemiological studies have yielded mixed results. While some have reported a positive effect of temperature on sea lice populations⁸⁵, others

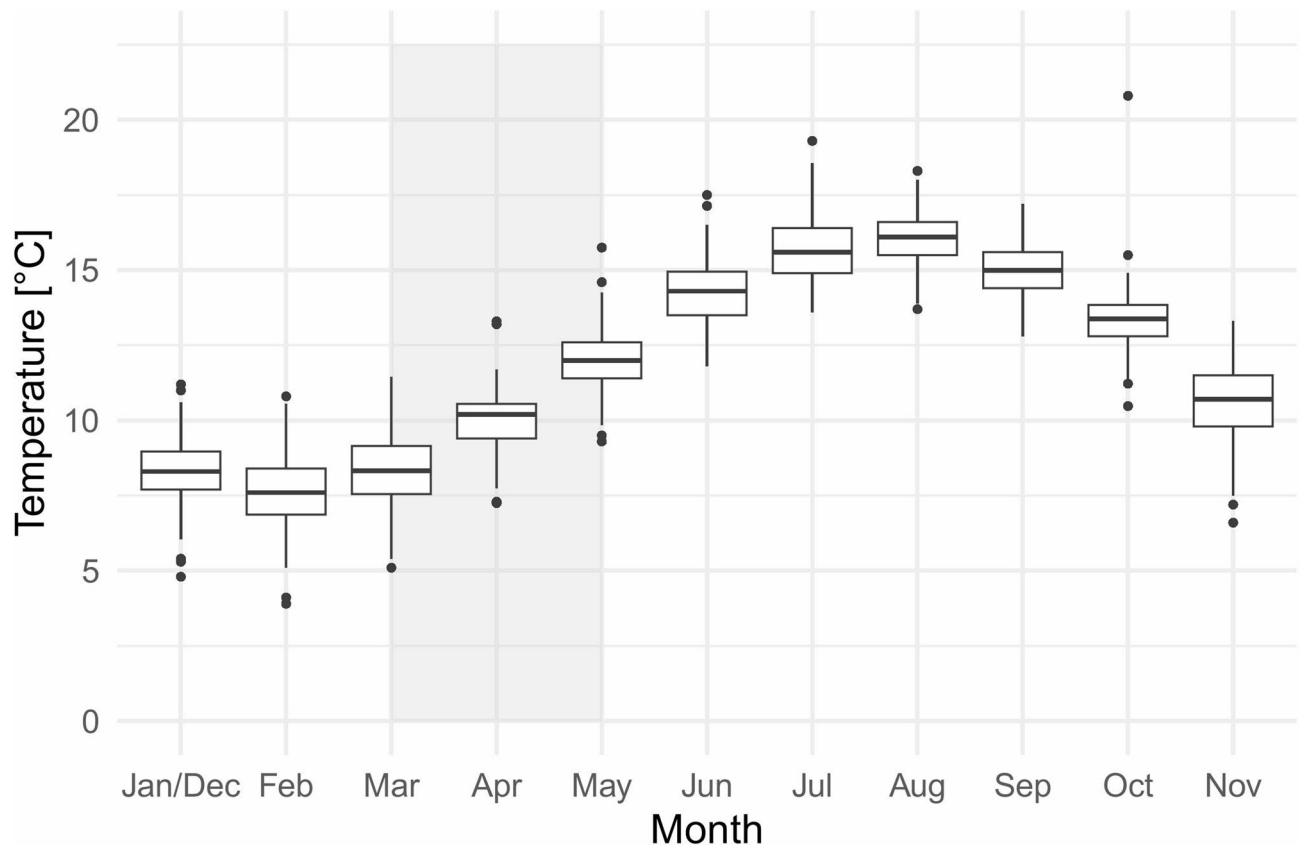


Fig. 5. Monthly surface seawater temperature in Irish bays from 2006 to 2023^{41–44}. Stricter management strategies implemented on salmon farms during the spring period (March to May)¹⁷ are highlighted in grey.

have found no significant impact and have suggested that temperature does not influence sea lice abundance at the population level^{3,19}.

Our findings on seasonal changes in sea lice abundance differed between the 30-year and 18-year datasets. In the 30-year dataset, sea lice abundance varied significantly across seasons. However, in the 18-year dataset, including additional predictors such as temperature, month was no longer a significant factor. Furthermore, we observed a negative effect of temperature on sea lice abundance. These results suggest that while seasonal patterns exist, they are likely driven more by broader variation in temperatures and other factors such as various management strategies than by season itself.

In the 30-year dataset, sea lice abundance declined in early spring, reaching the lowest levels observed in spring and early summer. The absence of a temperature-driven increase in sea lice abundance may be attributed to effective management strategies. Preventative treatments are implemented at the start of spring to maintain low lice levels, and overall abundance is kept substantially lower during this period compared to the rest of the year¹⁷. To protect out-migrating wild salmonid smolts, treatment trigger levels are reduced to 0.5 ovigerous females per fish in spring, compared to two ovigerous females per fish for the rest of the year¹⁷.

The strict management strategies implemented in spring not only reduce sea lice abundance during that period but may also delay the seasonal peak until autumn. By autumn, these strategies become less stringent, while seawater temperatures remain high (Fig. 5). As a result, the natural temperature-driven increase in sea lice levels is disrupted, which may explain the negative temperature effect we observed. Additionally, a lower sea lice dispersion rate was observed in higher temperatures⁸⁶ and the relationship between wild migrating salmonids and sea lice abundance during autumn has been discussed^{28,87}.

A similar seasonal pattern has been observed in Scotland and Canada, where sea lice levels remained low in spring but increase in autumn^{88,89}. The low spring abundance is likely due to strict management measures, which may obscure the natural seasonal variation in sea lice levels. In contrast, the rise in autumn has been linked to high seawater temperatures^{28,88–90} and higher incidence of other salmon pathogens connected with lower effectiveness of treatment^{91–94}.

Changes in sea lice abundance with farm exposure

Changes in sea lice abundance were observed in relation to the geographical factors of bay length, bay mouth width and number of active farms per bay, and the environmental variable dissolved oxygen. Collectively, these four factors likely reflect an impact of farm exposure, which appeared to influence the sea lice population dynamics analysed.

Numerous studies have examined the effect of farm location on sea lice abundance. To characterise farm locations, previous research considered oceanographic (e.g., current speed, flushing time, current direction), geographic (e.g., farm exposure, water depth, proximity to neighbouring farms), and environmental (e.g., salinity) factors^{3,4,19,22,83}. Findings indicate that long, narrow, and sheltered bays tend to have higher sea lice abundance due to reduced water circulation and increased sea lice retention. In contrast, shorter and wider bays promote better water exchange, leading to lower sea lice retention and abundance⁸³. Additionally, it was described that water current speed negatively correlates with sea lice abundance, while flushing time has a positive correlation, further supporting the suggestion that sheltered bays retain more sea lice¹⁹. Murray & Hall⁹⁵ reported a significant effect of bay's flushing time on treatment rates, although this relationship was influenced by regional oceanographic processes. However, Revie et al.³ suggested that coastal exposure at the farm level does not significantly impact sea lice abundance.

In this study, the width of the mouth of Irish coastal bays was found to have a negative relationship with sea lice abundance with wider bays having lower sea lice abundance. These findings align with previous research showing that sheltered bays, with narrower openings, have higher sea lice retention and, consequently, greater abundance^{19,83}. Here we also found that bay length was negatively related to sea lice abundance, with longer bays having lower sea lice abundance. In Ireland, longer bays do not necessarily contain more sheltered farms than shorter bays and since the longest bays in Ireland are also wider, higher sea lice retention is more strongly associated with shorter, narrower bays than longer bays. Additionally, dissolved oxygen levels were negatively correlated with sea lice abundance. Higher dissolved oxygen levels indicate greater water turbulence caused by waves and wind⁹⁶, which likely reduces sea lice retention in the water and decreases their ability to remain attached to their hosts^{97,98}.

In addition to the effect of bay geography, we found a significant positive relationship between the number of active salmon farms per bay and sea lice abundance. The dispersal of sea lice has been extensively studied^{31,86,99}, with several studies demonstrating transmission between farms^{4,86,100}. These findings are consistent with our observation that a higher number of salmon farms within the same bay is associated with increased sea lice abundance. Interestingly, Revie et al.¹⁹ identified only the presence of “bad neighboring farms” (those not following synchronized management practices) within 5 km as a predictor of interest, and even then, the effect was not statistically significant. To better understand the influence of neighboring farms on sea lice abundance, further analyses incorporating farm proximity and additional factors influencing sea lice dispersal—such as detailed bay topography, prevailing winds and currents, and management practices—would be valuable.

Conclusion

An extensive 30-year dataset was analysed to examine the effects of temporal, geographical, and environmental factors on sea lice abundance in Irish salmon farms. Our findings highlight seasonal variation and an influence of farm location; however, additional factors not included in this analysis, such as evolving husbandry practices and management strategies, may have influenced sea lice trends. Future research should incorporate these factors alongside variables like water currents, freshwater input, and proximity to other farms, to provide a more comprehensive understanding of sea lice dynamics. Additionally, analysing the abundance of different life stages could offer deeper insights into population fluctuations. Given the complex interplay of farm management, husbandry practices, ecological processes, and other contributing factors, long-term, multifaceted studies are essential. Continued research will be crucial in adapting to environmental changes and refining management strategies to mitigate sea lice impacts on salmon.

Data availability

The data used in this study is available at 10.6084/m9.figshare.30050932. The R code used in this study is available on GitHub at https://github.com/Klimesobe/Temporal_Geographical_Environmental_Effects_Seallice.git.

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Author contributions

B. K.: Conceptualization, Methodology, Validation, Formal analysis, Writing—Original Draft, Writing—Review & Editing, Visualization; O. L.: Methodology, Validation, Writing—Review & Editing; N. M. R.: Conceptualization, Writing—Review & Editing; J. D.: Data Curation, Writing—Review & Editing; A. T.: Writing—Review & Editing, Project administration, Funding acquisition; H. R.: Writing—Review & Editing; K. O.: Conceptualization, Methodology, Validation, Writing—Original Draft, Writing—Review & Editing, Supervision, Project administration. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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